



18 measured voices, short and long prompt dimensions

TTS LATENCY BENCHMARK ACROSS MODEL FAMILIES

This benchmark compares representative TTS voices available through Telnyx across short and long prompts. First audio latency shows how quickly a voice starts speaking in live voice experiences. RTF gives a normalized view of longer response generation.

FASTEST FIRST AUDIO P50

302 ms

Amazon Polly Standard, Joanna had the lowest first audio p50 in this run. Every measured voice completed 6/6 runs.

WEBSOCKET

Polly Standard, Joanna

302 ms

first audio p50

WEBSOCKET

Polly Neural, Joanna

310 ms

first audio p50

WEBSOCKET

Azure Neural, Jenny

327 ms

first audio p50

WEBSOCKET

Rime Coda, astra

340 ms

first audio p50

WEBSOCKET

xAI TTS, Eve

408 ms

first audio p50

WEBSOCKET

Minimax 2.8, Radiant Girl

413 ms

first audio p50

Polly Standard, Joanna

websocket

Polly Neural, Joanna

websocket

Azure Neural, Jenny

websocket

Rime Coda, astra

websocket

xAI TTS, Eve

websocket

Minimax 2.8, Radiant Girl

websocket

Inworld Mini, Abby

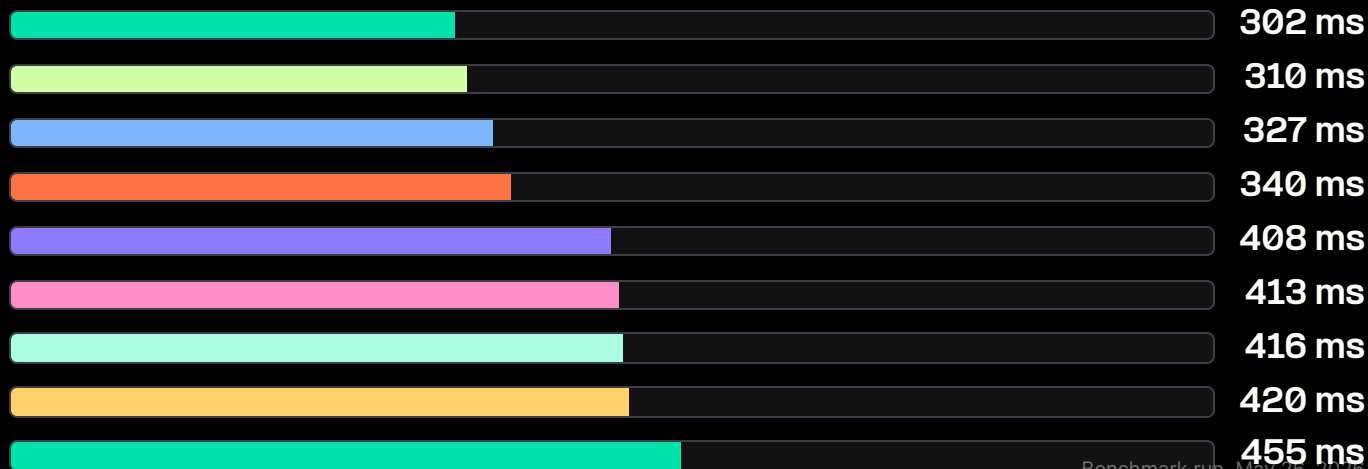
websocket

Rime Mist V2, ana

rest

Telnyx NaturalHD, astra

websocket



Prompt set: three short prompts plus one 127-word prompt

Benchmark run, May 26, 2026

FIRST AUDIO AND SYNTHESIS EFFICIENCY

First audio shows how quickly a voice starts speaking. RTF shows synthesis efficiency for longer responses, normalized by generated audio duration. Lower is better for both views.

First audio p50



RTF p50



WHAT THE RUN SAYS

All measured voices returned successful audio for 6/6 short-prompt runs. First audio is the clearest cross-provider latency signal for live voice agents. RTF is included so teams can compare synthesis efficiency without over-weighting clip length or voice speed.

FASTEST START

Polly Standard leads first audio

Amazon Polly Standard, Joanna finished at 302 ms first audio p50.

MEASURED FAMILY

Rime Coda measured

Rime Coda, astra finished at 340 ms first audio p50 and 0.37 RTF p50 in this run.

INTERPRETATION NOTE

Compare normalized output

Some providers return audio in larger chunks or at different speaking speeds, so RTF is the cleaner long-response efficiency metric.

MODEL FAMILY AND VOICE	INTERFACE	FIRST AUDIO P50	FIRST AUDIO P95	RTF P50	RTF P95	SUCCESS
Amazon Polly Standard, Joanna	websocket	302 ms	335 ms	0.08	0.09	6/6
Amazon Polly Neural, Joanna	websocket	310 ms	580 ms	0.08	0.17	6/6
Azure Neural, Jenny	websocket	327 ms	629 ms	0.09	0.12	6/6
Rime Coda, astra	websocket	340 ms	414 ms	0.37	0.41	6/6
xAI TTS, Eve	websocket	408 ms	1037 ms	0.11	0.56	6/6
Minimax 2.8 Turbo, Radiant Girl	websocket	413 ms	686 ms	0.11	0.23	6/6
Inworld Mini, Abby	websocket	416 ms	567 ms	0.15	0.21	6/6
Rime Mist V2, ana	rest	420 ms	443 ms	0.18	0.21	6/6
Telnyx NaturalHD, astra	websocket	455 ms	688 ms	0.14	0.47	6/6
Resemble Turbo, Alex	rest	458 ms	618 ms	0.16	0.32	6/6
Rime ArcanaV3, astra	websocket	473 ms	844 ms	0.12	0.63	6/6
Qwen3TTS, Delta	websocket	491 ms	589 ms	0.30	0.48	6/6
Inworld Max, Abby	websocket	493 ms	616 ms	0.28	0.31	6/6
Telnyx Natural, ana	rest	511 ms	632 ms	0.20	0.29	6/6
Azure Multilingual Neural, Ava	websocket	548 ms	637 ms	0.16	0.32	6/6
Azure Dragon HD, Ava	websocket	589 ms	786 ms	0.45	0.54	6/6
Telnyx Ultra, a7a591	rest	652 ms	850 ms	0.27	0.54	6/6
Resemble Pro, Aaron	rest	817 ms	832 ms	0.39	0.41	6/6

LONG TEXT CHANGES FIRST AUDIO BEHAVIOR

The benchmark includes a 127-word prompt as a separate dimension. Longer responses expose delivery behavior that short acknowledgement prompts can hide.

FASTEST LONG PROMPT

Inworld Mini, Abby

452 ms first audio p50 on the 127-word prompt.

POLLY STANDARD

1183 ms

+881 ms versus the short-prompt p50.

POLLY NEURAL

1286 ms

+975 ms versus the short-prompt p50.

Long prompt first audio p50

Top 10 by long-prompt first audio p50.



Short versus long prompt readout

MODEL FAMILY AND VOICE	SHORT P50	LONG P50	CHANGE	READOUT
Inworld Mini, Abby	416 ms	452 ms	+35 ms	Streaming path
Telnyx NaturalHD, astra	455 ms	477 ms	+22 ms	Streaming path
Rime ArcanaV3, astra	473 ms	538 ms	+65 ms	Streaming path
Rime Coda, astra	340 ms	578 ms	+239 ms	Streaming path
Minimax 2.8, Radiant Girl	413 ms	567 ms	+154 ms	Streaming path
Azure Neural, Jenny	327 ms	637 ms	+310 ms	Streaming path
Polly Standard, Joanna	302 ms	1183 ms	+881 ms	Sync response path
Polly Neural, Joanna	310 ms	1286 ms	+975 ms	Sync response path
Rime Mist V2, ana	420 ms	1157 ms	+736 ms	REST path
xAI TTS, Eve	408 ms	1100 ms	+692 ms	Streaming path
Resemble Turbo, Alex	458 ms	1126 ms	+668 ms	REST path
Telnyx Ultra, a7a591	652 ms	815 ms	+163 ms	REST path
Telnyx Natural, ana	511 ms	1071 ms	+560 ms	REST path

STREAMING PATH

Start can stay early

Several streaming providers still produced first audio under 650 ms on the long prompt.

SYNC PATH

Length matters more

Polly moved close to 1.2 to 1.3 seconds because first audio arrives with the completed clip.

PRACTICAL READOUT

Use both dimensions

Short prompts show how quickly speech starts. Long prompts show how each option handles fuller responses.

COVERAGE MAP

This page separates Telnyx product-family rows from direct provider/model rows. Each measured row uses the voice shown in the table.

MODEL FAMILY	VOICE TESTED	STATUS	PATH
Minimax 2.8 Turbo	Minimax 2.8 Turbo, Radiant Girl	Measured	WebSocket
Rime ArcanaV3	Rime ArcanaV3, astra	Measured	WebSocket
Telnyx NaturalHD	Telnyx NaturalHD, astra	Measured	WebSocket
Rime Coda	Rime Coda, astra	Measured	WebSocket
Rime Mist V2	Rime Mist V2, ana	Measured	REST
Telnyx Natural	Telnyx Natural, ana	Measured	REST
Inworld Mini	Inworld Mini, Abby	Measured	WebSocket
Inworld Max	Inworld Max, Abby	Measured	WebSocket
Qwen3TTS	Qwen3TTS, Delta	Measured	WebSocket
xAI TTS	xAI TTS, Eve	Measured	WebSocket
Resemble	Resemble Turbo and Pro	Measured	REST
Amazon Polly Standard	Amazon Polly Standard, Joanna	Measured	WebSocket
Amazon Polly Neural	Amazon Polly Neural, Joanna	Measured	WebSocket
Azure Neural	Azure Neural, Jenny	Measured	WebSocket
Azure Multilingual Neural	Azure Multilingual Neural, Ava	Measured	WebSocket
Azure Dragon HD	Azure Dragon HD, Ava	Measured	WebSocket
Telnyx Ultra	Telnyx Ultra, a7a591	Measured	REST
Google Neural	Not included in this run	Not measured	Test separately
Google Chirp 3 HD	Not included in this run	Not measured	Test separately
Azure Standard	Not included in this run	Not measured	Test separately

HOW TO READ THIS BENCHMARK

The benchmark sends the same prompts to each voice and records when audio first starts. Long text is shown separately because some providers respond differently as the generated speech gets longer.

MAIN METRIC

First audio p50

Median time from sending text to receiving first audio bytes.

SYNTHESIS EFFICIENCY

RTF p50

Median synthesis time divided by generated audio duration.

NOT RANKED

Final audio

Kept out of the ranking because chunking and voice speed differ by provider.

TRANSPORT

Interface

WebSocket for most providers. REST for Natural and Resemble in this run.

RUN COUNT

6 short, 2 long

Three short prompts with two runs each, plus one 127-word prompt with two runs.

COVERAGE NOTE

Outside this run

Some available families were outside this run and should be tested separately for customer-specific evaluations.

WHAT THIS MEANS FOR USE CASES

Use this as a directional latency readout. The best fit still depends on voice quality, language support, cloning needs, transport, and playback strategy.

LIVE VOICE AGENTS

Watch first audio

This is the delay users feel before the assistant starts speaking.

LONG CLIPS

Watch RTF

RTF is the better normalized synthesis metric for longer generated responses.

REST CAVEAT

Separate transport

REST responses can stream chunks, but they are not the same path as interactive WebSocket text streaming.

PROVIDER CAVEAT

Single chunk

Polly and some others can return all audio at once, so final audio equals first audio.

BUYING NOTE

Validate the fit

Use the same test with the customer's target voice, prompts, language, and region before making a final choice.

FAIR COMPARISON

Keep inputs fixed

Use the same region, prompts, endpoint, output format, and target voices when comparing options.